

Business Context

Reported videos can create a large review workload. This case study investigates whether available video metadata, engagement measures, author information, and transcription text can be used to distinguish claims from opinions and potentially support review prioritization.

Data

The source dataset contains 19,382 observations and 12 columns. The initial quality review identified 298 incomplete records affecting the target, transcription text, and engagement variables. These records were removed, leaving 19,084 observations. No duplicate records were found after cleaning. The target classes were relatively balanced: 9,608 claims and 9,476 opinions.

Analytical Approach

The workflow combined exploratory analysis, statistical testing, logistic regression, NLP feature engineering, and supervised machine learning.

A Welch two-sample t-test examined whether verified and unverified accounts had different mean video views. The test returned a p-value of approximately 2.61×10^{-120} . Mean views were approximately 265,664 for unverified accounts and 91,439 for verified accounts. This is statistical evidence of an association in this dataset, not evidence of causation.

For the main classification task, transcription text was converted into numerical features using CountVectorizer with unigrams and bigrams. The vectorizer learned 4,014 features from the training text only. The data was split into training, validation, and test sets, and model selection was kept separate from final test evaluation.

Model Comparison

Two ensemble classifiers were evaluated: Random Forest and XGBoost. Random Forest achieved cross-validated accuracy of 99.93%, precision of 100.00%, recall of 99.86%, and F1 of 99.93%. XGBoost achieved 99.84% accuracy, 99.97% precision, 99.72% recall, and 99.84% F1.

Random Forest was selected using validation recall because identifying actual claims was the primary modeling objective. On the validation set it achieved 99.95% recall, 100.00% precision, and 99.97% F1.

Final Test Result

The selected Random Forest was evaluated once on the untouched test set of 3,817 videos.

Accuracy: 99.95%

Precision: 100.00%

Recall: 99.90%

F1 Score: 99.95%

The confusion matrix shows 1,895 opinion videos correctly classified as opinions, 1,920 claim videos correctly classified as claims, 0 false positives, and 2 false negatives.

Key Insight

Feature importance indicates that video engagement variables, especially video views and likes, dominate the model's predictive signal. Text features have lower individual importance in the fitted Random Forest. These are model-based signals and should not be interpreted as causal effects.

Business Interpretation

Within this educational dataset, the model can distinguish the two content classes with very high predictive performance. A similar workflow could conceptually support prioritization of reported content for human review.

Limitations And Next Steps

The dataset was created for educational purposes and should not be treated as evidence about real TikTok moderation systems. Engagement features may also become available only after a video accumulates interactions, so feature availability at prediction time would need to be verified. Before any production use, the model should be tested on independent real-world data and monitored for changes in feature distributions. Additional signals such as user report history and author-level reporting patterns could also be investigated.